Well-founded recursion gives a function whose value depends on the values at smaller arguments

The Univalent Foundations Program

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Editorial note. The counted statement is the author's ‘thm:wfrec‘ (pinned-original-source.tex lines 1009-1016, original Lemma 10.3.7); the proof environment opens at line 1019, so the pairing is by adjacency.

Editorial note. The retained counter values (chapter 10, section 3, theorem offset 6) make the counted lemma print as Lemma 10.3.7, the number the first-edition label-number dump gives for ‘thm:wfrec‘.

Editorial note. The author's notational aside at source lines 1017-1018 (‘We are using the notation for images of subsets from \cref{sec:image}‘) lies between the two retained environments; it is presentational only and is not part of the statement or the proof, and the forward pointer it carries is deliberately outside the retained range.

Editorial note. No retained line refers to a label, citation or figure outside the retained span, so the package declares no external reference.

Editorial scope. One original result of the pinned first-edition chapter, counted once: the well-founded recursion principle for a set-valued function (author label thm:wfrec).

Allowed standard prerequisites (editorial).

- Well-founded relation and accessibility. A relation $<$ on A is well-founded when every a:A is accessible, written acc(a); acc(a) is the type of witnesses that all predecessors of a are accessible.. It is the hypothesis of the statement and the object the proof recurses on.
- Power set. power B is the type B->prop, the type of subsets of the set B.. The recursion datum g is a function power B -> B.
- Image of a subset. setof{ f(a') | a'<a } is the subset of B consisting of the values of f on the predecessors of a.. It is the argument at which g is evaluated in the recursion equation.

Lemma 10.3.7. *Suppose B is a set and we have a function*

$$g : \mathcal{P}(B) \to B$$

Then if $<$ is a well-founded relation on A, there is a function $f : A \to B$ such that for all $a : A$ we have

$$f(a) = g\Big(\{\, f(a') \mid a' < a \,\} \Big).$$

Proof. We first define, for every $a : A$ and $s : \mathsf{acc}(a)$, an element $\bar{f}(a, s) : B$. By induction, it suffices to assume that s is a function assigning to each $a' < a$ a witness $s(a') : \mathsf{acc}(a')$, and that moreover for each such a' we have an element $\bar{f}(a', s(a')) : B$. In this case, we define

$$\bar{f}(a, s) :\equiv g\Big(\{\ \bar{f}(a', s(a')) \mid a' < a\ \} \Big).$$

Now since $<$ is well-founded, we have a function $w : \prod_{(a:A)} \mathsf{acc}(a)$. Thus, we can define $f(a) :\equiv \bar{f}(a, w(a))$. □